

IMPORT DATA USING TRANSFORM MAP (SPREADSHEET)

Project Documentation

1. Introduction

Item	Details
Project Title	Import Data Using Transform Map (Spreadsheet)
Platform	Developer ServiceNow
Project Type	Data Import and Transformation
Team Members	Monika T S -Team Lead Nakshtraa ST-Developer Sadhana K - Tester&Coordinator

2. Project Overview

Purpose

Import employee data from an Excel spreadsheet into ServiceNow using an Import Set and Transform Map. The process maps source columns to target fields, validates data, and inserts or updates records according to the configured transformation rules.

Key Features

- Upload Excel (.xlsx) or CSV files as a data source.
- Create Import Set Tables to stage source data.
- Configure Transform Maps and source-to-target field mappings.
- Apply transformation rules and validate data formats.
- Handle duplicate records using configured coalesce fields.
- Review import results, transformation history, and errors.

3. Architecture

The solution uses a three-stage data flow: spreadsheet source → Import Set Table → Transform Map → target table.

Component	Description
Source	Employee_Data.xlsx containing fields such as First Name, Last Name, Email, and Department.
Import Set Table	Temporary staging table that stores rows loaded from the spreadsheet.
Transform Map	Defines field mappings and transformation behavior from the import set to the target table.
Target Table	ServiceNow User table (sys_user), subject to the instance's schema and access permissions.

4. Setup Instructions

Prerequisites

- An active ServiceNow instance and authorized account.
- Permissions to load data, create Transform Maps, and access the target table.
- A prepared Excel or CSV file with consistent column headers.
- A supported web browser.

Configuration Steps

1. Sign in to the ServiceNow instance.
2. Navigate to System Import Sets → Load Data.
3. Select Employee_Data.xlsx and configure or create the Import Set Table.
4. Review the loaded source rows and confirm column headers.
5. Create a Transform Map and select the appropriate target table.
6. Configure source-to-target field maps and any required transformation logic.
7. Configure a coalesce field if matching existing records should be updated rather than duplicated.
8. Run the Transform Map and review the execution results.
9. Verify the resulting records in the target table.

Example Source Spreadsheet

First Name	Last Name	Email	Department
Arun	Kumar	arun@example.com	IT
Priya	Sharma	priya@example.com	HR
Rahul	Das	rahul@example.com	Finance

5. Folder / Artifact Structure

This is a ServiceNow platform configuration project and does not require a conventional React or Node.js source-code directory. The following logical organization can be used for project deliverables:

```
Import_Data_Transform_Map/  
├── Source_Data/  
│   └── Employee_Data.xlsx  
├── Import_Set/  
│   └── Employee_Import_Set_Table  
├── Transform_Map/  
│   └── Employee_Data_Transform_Map  
├── Test_Cases/  
│   └── Import_Test_Cases  
└── Documentation/  
    └── Project_Documentation
```

6. Running the Import

10. Open the ServiceNow instance and navigate to Load Data.
11. Upload the spreadsheet and load it into an Import Set Table.
12. Open or create the Transform Map and verify field mappings.
13. Execute the transformation.
14. Review transform history, row counts, and error messages.
15. Confirm that target records contain the expected values.

Expected result: valid source rows are transformed and written to the configured target table; rejected or skipped rows are identifiable through import and transformation results.

7. Import Configuration Documentation

Configuration	Example Value
Source File	Employee_Data.xlsx
File Format	Excel (.xlsx)
Import Set Table	Employee Import Set Table
Transform Map	Employee Data Transform Map
Target Table	User (sys_user)
Import Method	Spreadsheet upload
Duplicate Handling	Coalesce on Email, if appropriate for the target data
Execution	Run Transform Map

Example Field Mapping

Spreadsheet Source Column	Target Field (example)
First Name	first_name
Last Name	last_name
Email	email
Department	department (confirm target reference-field requirements)
Phone Number	phone

Actual target fields and reference-field handling should be confirmed against the ServiceNow instance's table dictionary and project requirements.

8. Authentication and Authorization

Authentication is provided by the ServiceNow instance. Access to data loading, Transform Map configuration, and target records is controlled by assigned roles and ACLs. Use authorized test credentials and do not place passwords or secrets in the spreadsheet or documentation.

9. User Interface

- Load Data page – upload the source spreadsheet.
- Import Set Table – inspect staged source records.
- Transform Map form – configure the target table and transformation behavior.
- Field Maps – define source-to-target relationships.
- Transform History – review execution status and errors.
- Target Table – verify imported or updated records.

Add project-specific screenshots of the upload page, Import Set rows, Transform Map, field mappings, execution results, and target records here.

10. Testing

Testing includes functional verification of file loading, field mapping, transformation, duplicate handling, invalid data, and final target records.

Test Case ID	Test Scenario	Expected Result
TC-001	Upload a valid spreadsheet	File loads and source rows appear in the

		Import Set Table.
TC-002	Create an Import Set Table	The staging table is created or selected successfully.
TC-003	Configure the Transform Map	The map saves with the intended target table.
TC-004	Map source fields	Each source column maps to the correct target field.
TC-005	Run transformation	Valid records are inserted or updated as configured.
TC-006	Import duplicate data	Coalesce behavior prevents unintended duplicate records.
TC-007	Import invalid or missing data	Invalid rows are handled as configured and errors are traceable.
TC-008	Verify target records	Target values match the expected transformed data.

Suggested tools: ServiceNow Import Sets, Transform Maps, Transform History, target-table list views, and spreadsheet review.

11. Screenshots or Demo

16. Employee spreadsheet containing source data.
17. ServiceNow Load Data page with the uploaded file.
18. Import Set Table showing staged records.
19. Transform Map configuration.
20. Source-to-target field mappings.
21. Transform execution results and history.
22. Imported records in the target table.

Project Demonstration Video: <https://drive.google.com/file/d/1YyyuqXpJxw9ZX-8Z9ygG90i6guhO4ws6/view?usp=sharing>

12. Known Issues / Risks

- Incorrect field mapping can place values in unintended target fields.
- Missing mandatory values may cause records to fail or be rejected.
- Without suitable coalesce configuration, duplicate records may be created.
- Inconsistent spreadsheet headers or data formats can affect loading and transformation.
- Reference fields, such as Department, may require a valid reference value or specific mapping logic.
- Large spreadsheets may take longer to process.
- Insufficient roles or ACL permissions may prevent loading or record updates.

13. Future Enhancements

- Schedule recurring imports from a controlled data source.
- Add pre-import validation and clearer error reporting.
- Create reusable transformation logic for complex formats.
- Generate import summary reports and operational dashboards.
- Improve duplicate detection and reconciliation.
- Add automated regression tests for field mappings and data accuracy.

- Configure notifications for failed or completed imports.
- Extend the workflow to additional source files and target tables.

Project Outcome

The project demonstrates spreadsheet-based data migration into ServiceNow using Import Sets and Transform Maps. It provides a repeatable process for staging source data, mapping fields, applying transformation rules, and validating resulting records.